

Bandits on Graphs

Exploiting smoothness and side observations

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joint work with

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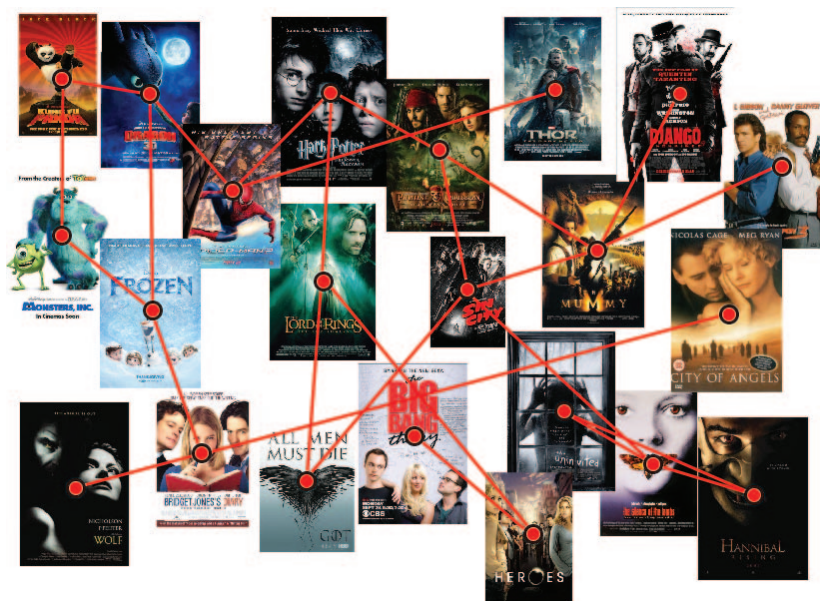
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Gergely Neu (SequeL INRIA)







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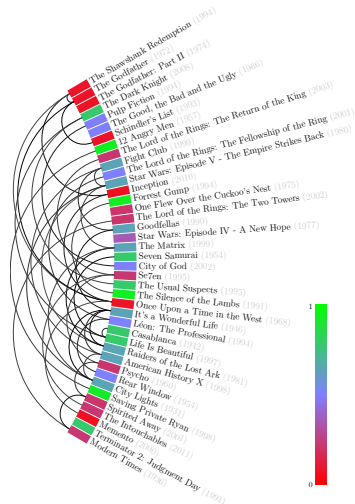
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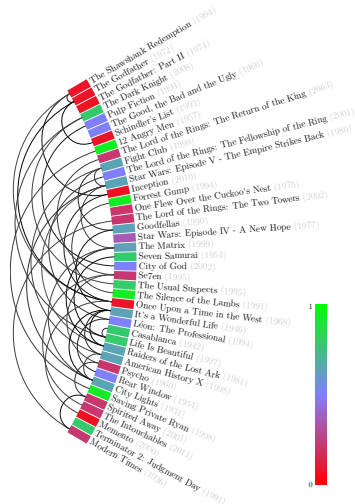
Movie recommendation: (in each time step)

- Recommend movies to a **single user**.



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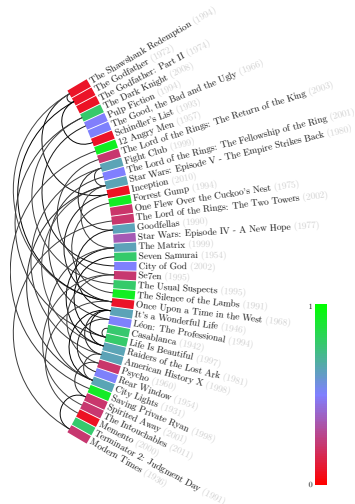


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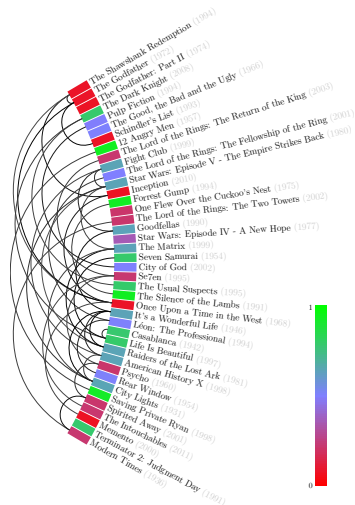
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Assumptions:

- ▶ Unknown reward function $f : V(G) \rightarrow \mathbb{R}$.
- ▶ Function f is **smooth** on a graph.
- ▶ Neighboring movies $\Rightarrow$ similar preferences.
- ▶ Similar preferences $\nRightarrow$ neighboring movies.



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Smoothness of the function:

$$S_G(f) = \frac{1}{2} \sum_{i,j \leq N} w_{i,j} (f_i - f_j)^2$$

Smaller value of $S_G(f)$, smoother the function f is.

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Examples:

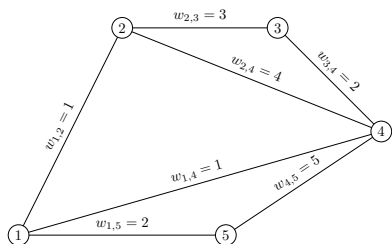
- ▶ **Complete graph:** Only constant function has smoothness 0.
- ▶ **Edgeless graph:** Every function has smoothness 0.
- ▶ **Constant function:** Smoothness 0 for every graph.

Graph Laplacian

- ▶ $\mathcal{W}$: $N \times N$ matrix of the edge weights $w_{i,j}$.
- ▶ $\mathcal{D}$: Diagonal matrix with the entries $d_i = \sum_j w_{i,j}$.
- ▶ $\mathcal{L} = \mathcal{D} - \mathcal{W}$: Graph Laplacian.
 - ▶ Positive semidefinite matrix.
 - ▶ Diagonally dominant matrix.

Example:

$$\mathcal{L} = \begin{pmatrix} 4 & -1 & 0 & -1 & -2 \\ -1 & 8 & -3 & -4 & 0 \\ 0 & -3 & 5 & -2 & 0 \\ -1 & -4 & -2 & 12 & -5 \\ -2 & 0 & 0 & -5 & 7 \end{pmatrix}$$



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Note: $\mathbf{Q}^\top \mathbf{f} = \alpha^*$

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$$S_G(f) = \mathbf{f}^\top \mathcal{L} \mathbf{f} = \mathbf{f}^\top \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{f} = \alpha^{*\top} \mathbf{\Lambda} \alpha^* = \|\alpha^*\|_{\mathbf{\Lambda}}^2 = \sum_{i=1}^N \lambda_i (\alpha_i^*)^2$$

Smoothness and regularization: Small value of

(a) $S_G(f)$ (b) $\mathbf{\Lambda}$ norm of α^* (c) α_i^* for large λ_i

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- ▶ In each time t step choose a node $\pi(t)$.
- ▶ the $\pi(t)$ -th row $\mathbf{x}_{\pi(t)}$ of the matrix $\mathbf{Q}$ corresponds to the arm $\pi(t)$.
- ▶ Obtain noisy reward $r_t = \mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^* + \varepsilon_t$. Note: $\mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^* = f_{\pi(t)}$
 - ▶ ε_t is R -sub-Gaussian noise. $\forall \xi \in \mathbb{R}, \mathbb{E}[e^{\xi \varepsilon_t}] \leq \exp(\xi^2 R^2 / 2)$
- ▶ Minimize cumulative regret

$$R_T = T \max_a (\mathbf{x}_a^\top \boldsymbol{\alpha}^*) - \sum_{t=1}^T \mathbf{x}_{\pi(t)}^\top \boldsymbol{\alpha}^*.$$

- ▶ Can't we just use *linear bandits*?

Solutions

- ▶ **Linear bandit algorithms**

(Existing solutions)

- ▶ **LinUCB**

(Li et al., 2010)

- ▶ Regret bound $\approx D\sqrt{T \ln T}$

- ▶ **LinearTS**

(Agrawal and Goyal, 2013)

- ▶ Regret bound $\approx D\sqrt{T \ln N}$

Note: D is ambient dimension, in our case N , length of x_i .

Number of actions, e.g., all possible movies → **HUGE!**

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Effective dimension

- ▶ **Effective dimension:** Largest d such that

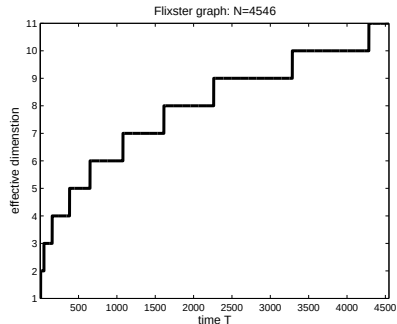
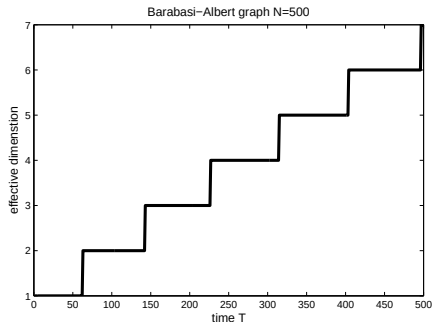
$$(d - 1)\lambda_d \leq \frac{T}{\log(1 + T/\lambda)}.$$

- ▶ Function of time horizon and graph properties
- ▶ λ_i : i -th smallest eigenvalue of $\mathbf{A}$.
- ▶ λ : Regularization parameter of the algorithm.

Properties:

- ▶ d is small when the coefficients λ_i grow rapidly above time.
- ▶ d is related to the number of “non-negligible” dimensions.
- ▶ Usually d is much smaller than D in real world graphs.
- ▶ Can be computed beforehand.

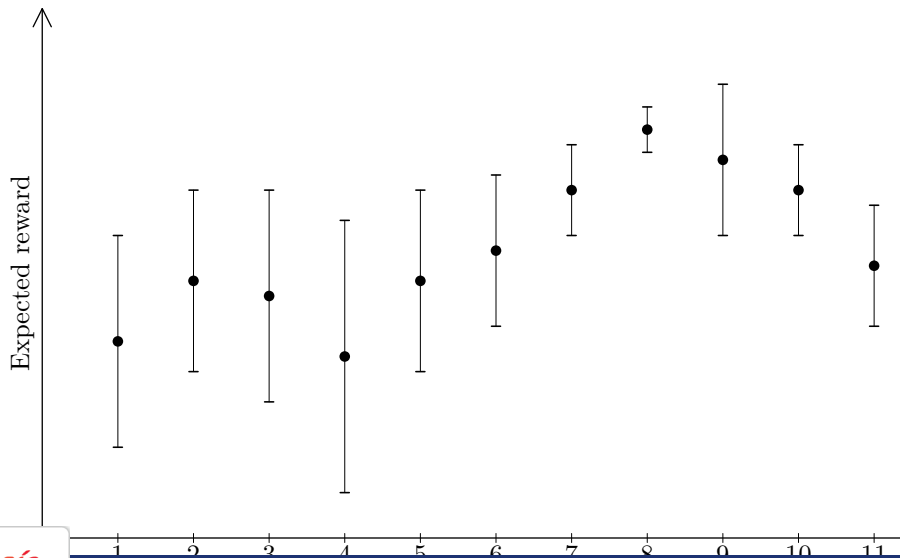
Effective dimension vs. Ambient dimension



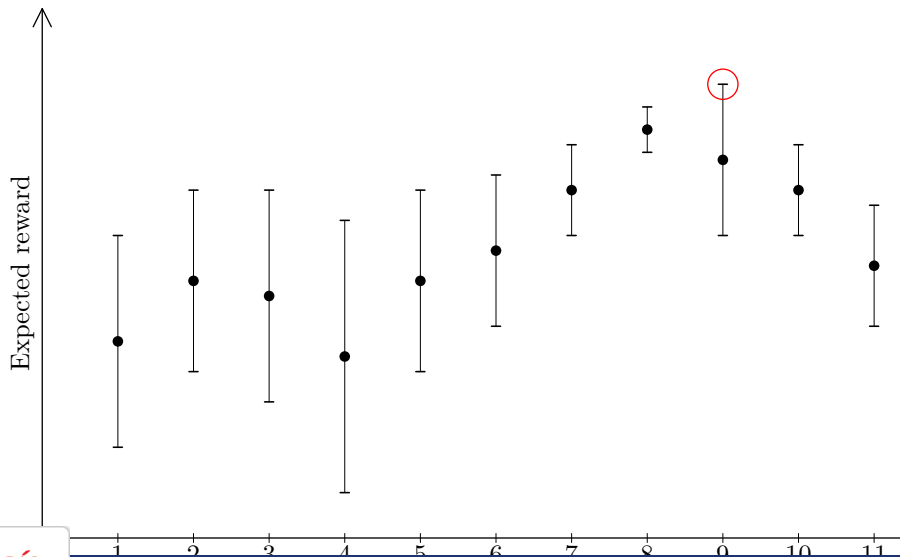
$$d \ll D$$

Note: In our setting $T < N = D$.

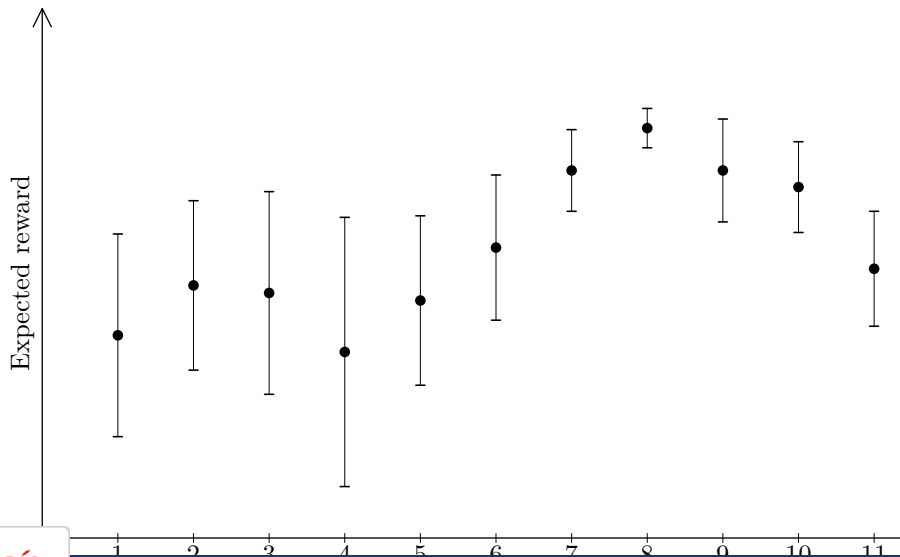
UCB style algorithms: **Estimate**



UCB style algorithms: **Sample**



UCB style algorithms: **Estimate** ...



SpectralUCB

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1: Input:
2:    $N, T, \{\mathbf{\Lambda}_{\mathcal{L}}, \mathbf{Q}\}, \lambda, \delta, R, C, \mathcal{L}$ 
3: Run:
4:    $\mathbf{\Lambda} \leftarrow \mathbf{\Lambda}_{\mathcal{L}} + \lambda \mathbf{I}$ 
5:    $d \leftarrow \max\{d : (d-1)\lambda_d \leq T/\ln(1+T/\lambda)\}$ 
6: for  $t = 1$  to  $T$  do
7:   Update the basis coefficients  $\hat{\alpha}$ :
8:    $\mathbf{X}_t \leftarrow [\mathbf{x}_{\pi(1)}, \dots, \mathbf{x}_{\pi(t-1)}]^\top$ 
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13:   $\pi(t) \leftarrow \arg \max_a \left( \mathbf{x}_a^\top \hat{\alpha} + c_t \|\mathbf{x}_a\|_{\mathbf{V}_t^{-1}} \right)$ 
14:  Observe the reward  $r_t$ 
15: end for

```

SpectralUCB regret bound

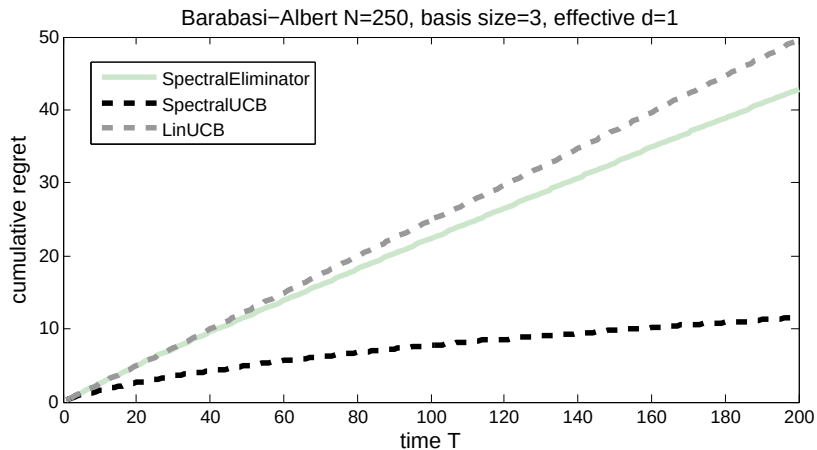
- ▶ d : Effective dimension.
- ▶ λ : Minimal eigenvalue of $\mathbf{\Lambda} = \mathbf{\Lambda}_{\mathcal{L}} + \lambda \mathbf{I}$.
- ▶ C : Smoothness upper bound, $\|\boldsymbol{\alpha}^*\|_{\mathbf{\Lambda}} \leq C$.
- ▶ $\mathbf{x}_i^\top \boldsymbol{\alpha}^* \in [-1, 1]$ for all i .

The **cumulative regret** R_T of **SpectralUCB** is with probability $1 - \delta$ bounded as

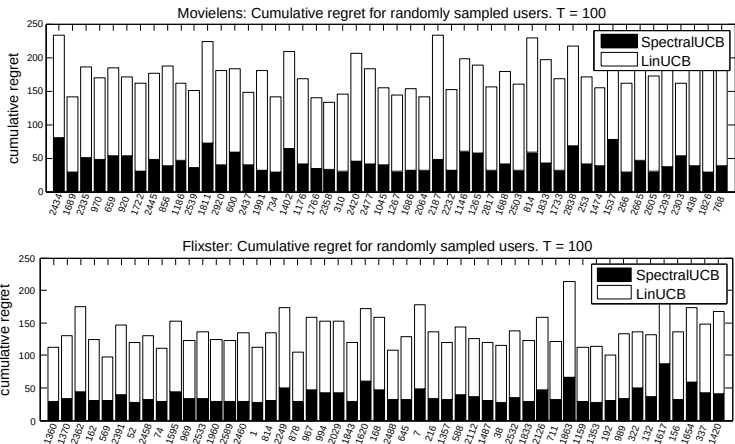
$$R_T \leq \left(8R \sqrt{d \ln \frac{\lambda + T}{\lambda} + 2 \ln \frac{1}{\delta} + 4C + 4} \right) \sqrt{dT \ln \frac{\lambda + T}{\lambda}}.$$

$$R_T \approx d \sqrt{T \ln T}$$

Synthetic experiment

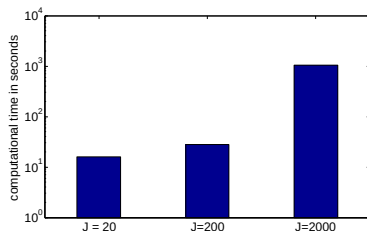
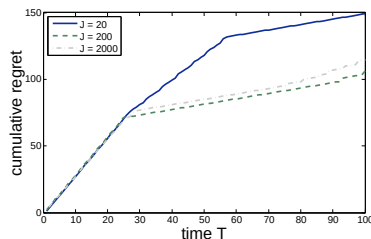


Real world experiment



Improving the running time: reduced eigenbasis

- ▶ **Reduced basis:** We only need first few eigenvectors.
- ▶ **Getting J eigenvectors:** $\mathcal{O}(Jm \log m)$ time for m edges
- ▶ Computationally less expensive, comparable performance.



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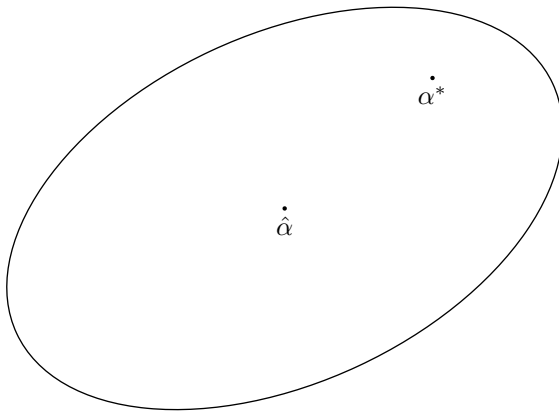
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 - ▶ Compute posterior distribution according to reward received

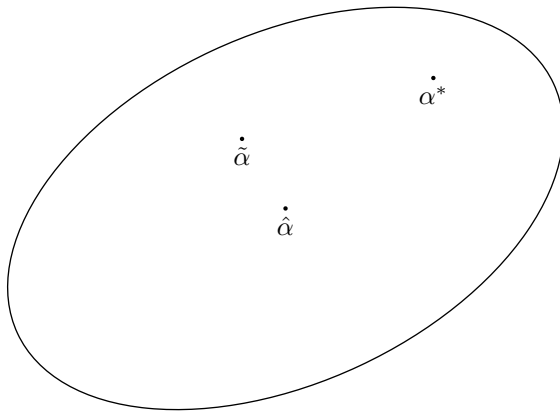
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- ▶ Only requires $D^2 + DN \rightarrow N^2$ per step update

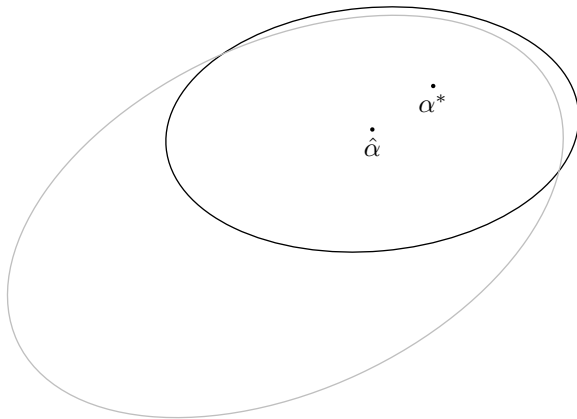
Thomson Sampling: **Estimate**



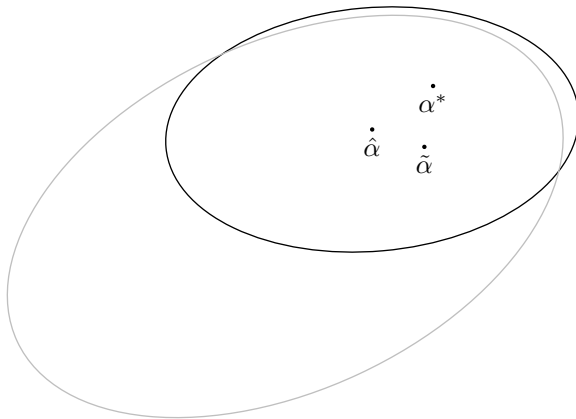
Thomson Sampling: **Sample**



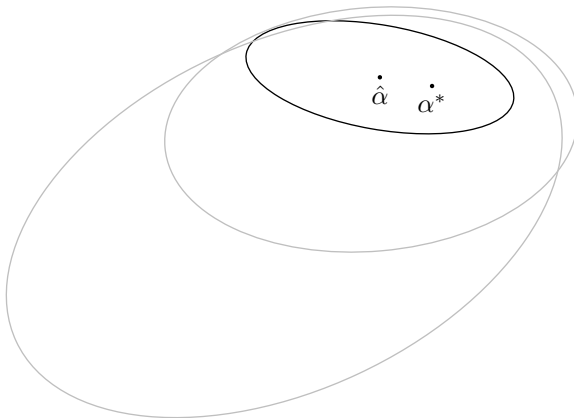
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Thomson Sampling: **Sample**



Thomson Sampling: **Estimate** ...



SpectralTS algorithm

- 1: **Input:**
- 2: $N, T, \{\mathbf{\Lambda}_{\mathcal{L}}, \mathbf{Q}\}, \lambda, \delta, R, C$
- 3: **Initialization:**
- 4: $v = R\sqrt{6d \log((\lambda + T)/\delta\lambda)} + C$
- 5: $\hat{\boldsymbol{\alpha}} = \mathbf{0}_N$
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- 8: **Run:**
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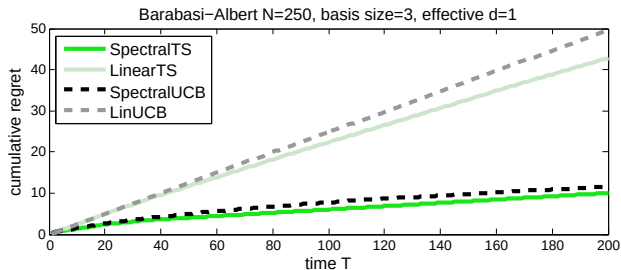
$$\mathcal{R}_T \leq \frac{11g}{p} \sqrt{\frac{4+4\lambda}{\lambda} dT \log \frac{\lambda+T}{\lambda}} + \frac{1}{T} + \frac{g}{p} \left(\frac{11}{\sqrt{\lambda}} + 2 \right) \sqrt{2T \log \frac{2}{\delta}},$$

where $p = 1/(4e\sqrt{\pi})$ and

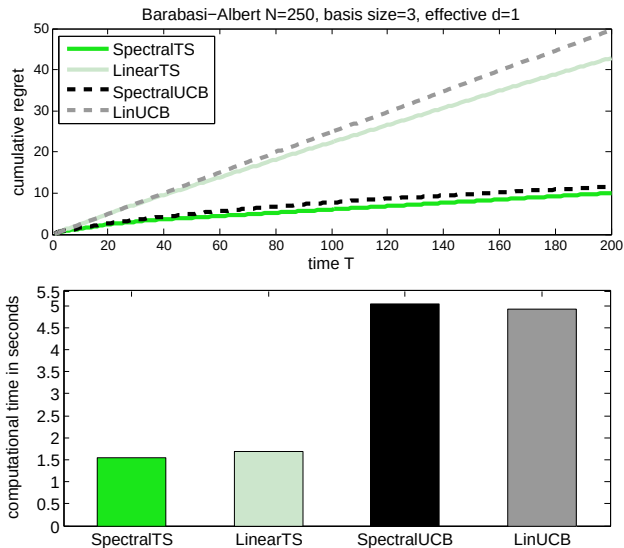
$$g = \sqrt{4 \log TN} \left(R \sqrt{6d \log \left(\frac{\lambda+T}{\delta\lambda} \right)} + C \right) + R \sqrt{2d \log \left(\frac{(\lambda+T)T^2}{\delta\lambda} \right)} + C.$$

$$R_T \approx d \sqrt{T \log N}$$

Synthetic experiment

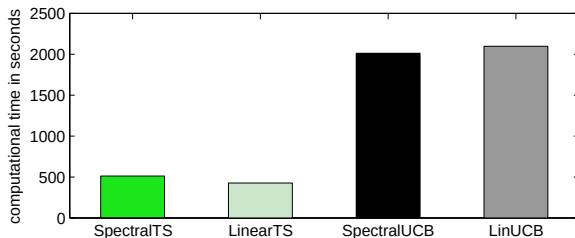
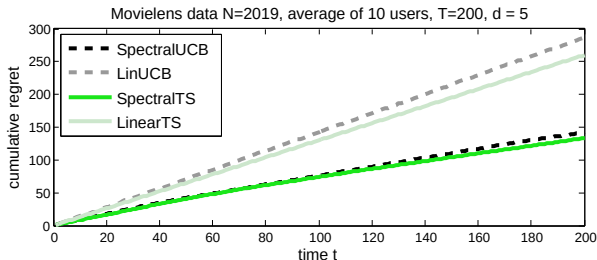


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Real world experiment

MovieLens dataset of 6k users who rated one million movies.



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Exploiting side observations

Example 1: undirected observations



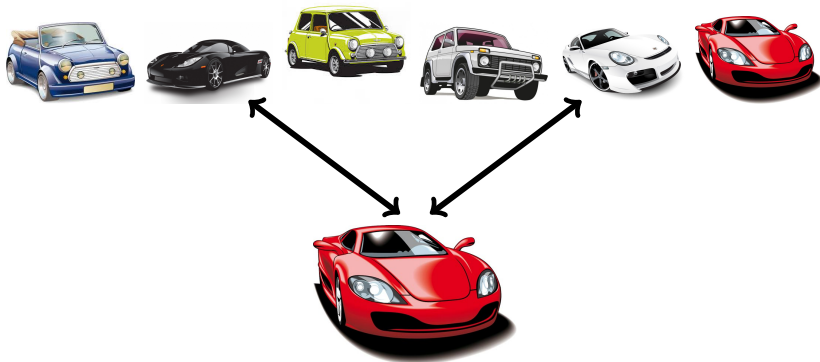
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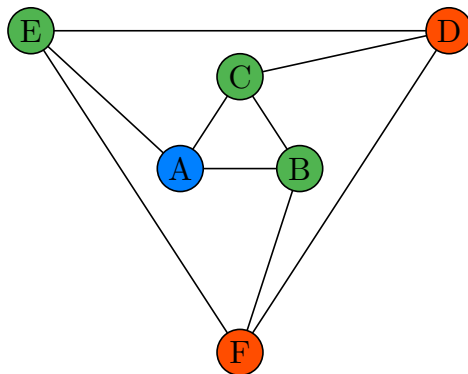


Exploiting side observations

Example 1: undirected observations



Example 1: Graph Representation



Example 2: Directed observation



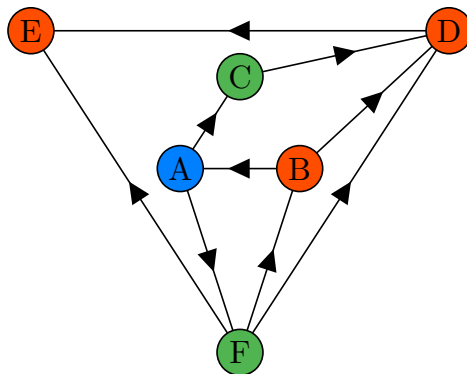
Example 2: Directed observation



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Example 2



Learning setting

In each time step $t = 1, \dots, T$

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- ▶ Plays action $I_t \in [N]$
- ▶ Obtain loss ℓ_{t,I_t} of action played
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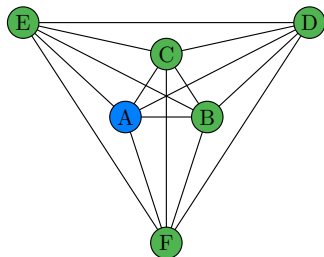
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► **Performance measure:** Total expected regret

$$R_T = \max_{i \in [N]} \mathbb{E} \left[\sum_{t=1}^T (\ell_{t,I_t} - \ell_{t,i}) \right]$$

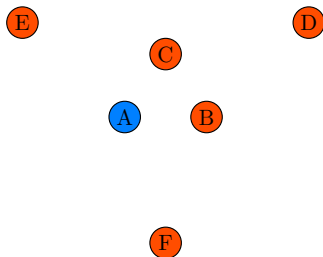
Full Information setting

- ▶ Pick an action (e.g. action A)
- ▶ Observe losses of all actions
- ▶ $R_T = \tilde{O}(\sqrt{T})$



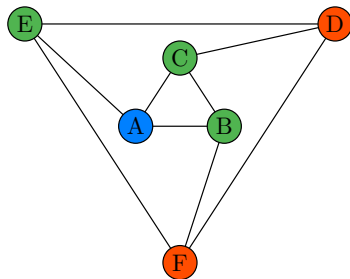
Bandit setting

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Side observation (Undirected case)

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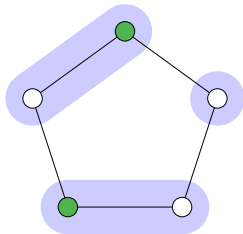
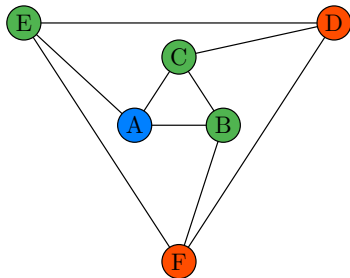


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Mannor and Shamir (ELP algorithm)

- ▶ Need to know graph
- ▶ Clique decomposition (c cliques)
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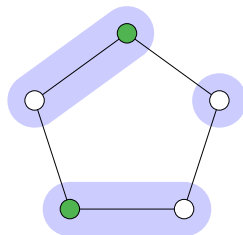
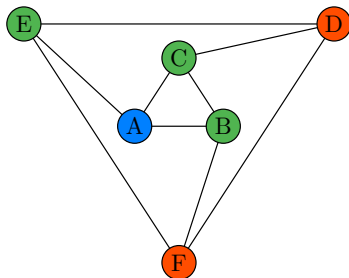
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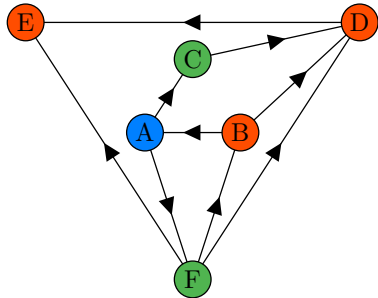
Alon, Cesa-Bianchi, Gentile, Mansour

- ▶ No need to know graph
- ▶ Independence set of α actions
- ▶ $R_T = \tilde{O}(\sqrt{\alpha T})$



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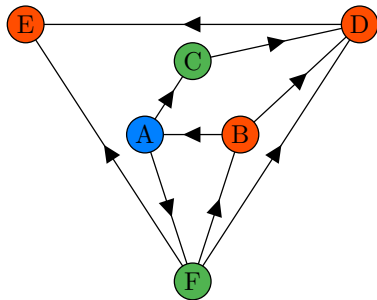


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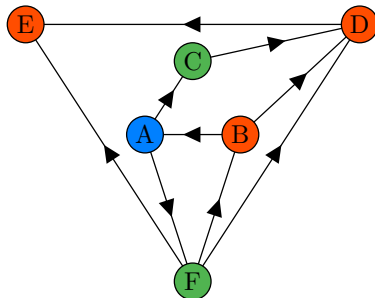
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Our solution: Exp3-IX

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Exp3 algorithms in general

- **Compute weights** using loss estimates $\hat{\ell}_{t,i}$.

$$w_{t,i} = \exp \left(-\eta \sum_{s=1}^{t-1} \hat{\ell}_{s,i} \right)$$

- **Play action** I_t such that

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$$\mathbb{P}(I_t = i) = p_{t,i} = \frac{w_{t,i}}{W_t} = \frac{w_{t,i}}{\sum_{j=1}^N w_{t,j}}$$

- **Update loss estimates** (using observability graph)

How the algorithms approach to bias variance tradeoff?

Bias variance tradeoff approaches

- ▶ Approach of previous algorithms – **Mixing**
 - ▶ Bias sampling distribution $\mathbf{p}_t$ over actions
 - ▶ $\mathbf{p}'_t = (1 - \gamma)\mathbf{p}_t + \gamma\mathbf{s}_t$ – mixed distribution
 - ▶ $\mathbf{s}_t$ – probability distribution which supports exploration
 - ▶ Loss estimates $\hat{\ell}_{t,i}$ are unbiased
- ▶ Approach of our algorithm – **Implicit eXploration (IX)**
 - ▶ Bias loss estimates $\hat{\ell}_{t,i}$
 - ▶ Biased loss estimates $\implies$ biased weights
 - ▶ Biased weights $\implies$ biased probability distribution
 - ▶ No need for mixing

Mannor and Shamir - ELP algorithm

- ▶ $\mathbb{E}[\hat{\ell}_{t,i}] = \ell_{t,i}$ – unbiased loss estimates
- ▶ $p'_{t,i} = (1 - \gamma)p_{t,i} + \gamma s_{t,i}$ – bias by mixing
- ▶ $\mathbf{s}_t = \{s_{t,1}, \dots, s_{t,N}\}$ – probability distribution over the action set

$$\mathbf{s}_t = \arg \max_{\mathbf{s}_t} \left[\min_{j \in [N]} \left(s_{t,j} + \sum_{k \in N_{t,j}} s_{t,k} \right) \right] = \arg \max_{\mathbf{s}_t} \left[\min_{j \in [N]} q_{t,j} \right]$$

- ▶ $q_{t,j}$ – probability that loss of j is observed according to $\mathbf{s}_t$

Mannor and Shamir - ELP algorithm

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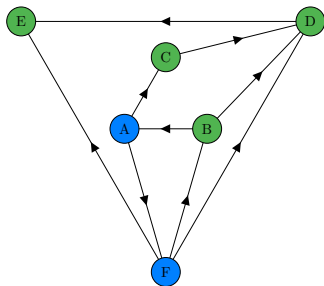
- ▶ $q_{t,j}$ – probability that loss of j is observed according to $\mathbf{s}_t$
- ▶ **Computation of $\mathbf{s}_t$**
 - ▶ Graph needs to be disclosed
 - ▶ Solving simple linear program
- ▶ Needs to know graph before playing an action
- ▶ Graphs can be only undirected

Alon, Cesa-Bianchi, Gentile, Mansour - Exp3-DOM

- ▶ $\mathbb{E}[\hat{\ell}_{t,i}] = \ell_{t,i}$ – unbiased loss estimates
- ▶ $p'_{t,i} = (1 - \gamma)p_{t,i} + \gamma s_{t,i}$ – bias by mixing
- ▶ $\mathbf{s}_t = \{s_{t,1}, \dots, s_{t,N}\}$ – probability distribution over the action set

$$s_{t,i} = \begin{cases} \frac{1}{r} & \text{if } i \in R; |R| = r \\ 0 & \text{otherwise.} \end{cases}$$

- ▶ R – dominating set of r elements
- ▶ $\mathbf{s}_t$ – uniform distribution over R
- ▶ Needs to know graph beforehand
- ▶ Graphs can be directed



Previous algorithms - loss estimates

$$\hat{\ell}_{t,i} = \begin{cases} \ell_{t,i}/o_{t,i} & \text{if } \ell_{t,i} \text{ is observed} \\ 0 & \text{otherwise.} \end{cases}$$

$$\mathbb{E}[\hat{\ell}_{t,i}] = \frac{\ell_{t,i}}{o_{t,i}} o_{t,i} + 0(1 - o_{t,i}) = \ell_{t,i}$$

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Exp3-IX - loss estimates

$$\hat{\ell}_{t,i} = \begin{cases} \ell_{t,i}/(o_{t,i} + \gamma) & \text{if } \ell_{t,i} \text{ is observed} \\ 0 & \text{otherwise.} \end{cases}$$

$$\mathbb{E}[\hat{\ell}_{t,i}] = \frac{\ell_{t,i}}{o_{t,i} + \gamma} o_{t,i} + 0(1 - o_{t,i}) = \ell_{t,i} - \ell_{t,i} \frac{\gamma}{o_{t,i} + \gamma} \leq \ell_{t,i}$$

► No mixing!

Analysis of Exp3 algorithms in general

- Evolution of W_{t+1}/W_t

$$\frac{1}{\eta} \log \frac{W_{t+1}}{W_t} = \frac{1}{\eta} \log \left(1 - \eta \sum_{i=1}^N p_{t,i} \hat{\ell}_{t,i} + \frac{\eta^2}{2} \sum_{i=1}^N p_{t,i} (\hat{\ell}_{t,i})^2 \right),$$

$$\sum_{i=1}^N p_{t,i} \hat{\ell}_{t,i} \leq \left[\frac{\log W_t}{\eta} - \frac{\log W_{t+1}}{\eta} \right] + \frac{\eta}{2} \sum_{i=1}^N p_{t,i} (\hat{\ell}_{t,i})^2$$

- Taking expectation and summing over time

$$\mathbb{E} \left[\sum_{t=1}^T \sum_{i=1}^N p_{t,i} \hat{\ell}_{t,i} \right] - \mathbb{E} \left[\sum_{t=1}^T \hat{\ell}_{t,k} \right] \leq \mathbb{E} \left[\frac{\log N}{\eta} \right] + \mathbb{E} \left[\frac{\eta}{2} \sum_{t=1}^T \sum_{i=1}^N p_{t,i} (\hat{\ell}_{t,i})^2 \right]$$

Regret bound of Exp3-IX

$$\underbrace{\mathbb{E} \left[\sum_{t=1}^T \sum_{i=1}^N p_{t,i} \hat{\ell}_{t,i} \right]}_A - \underbrace{\mathbb{E} \left[\sum_{t=1}^T \ell_{t,k} \right]}_B \leq \mathbb{E} \left[\frac{\log N}{\eta} \right] + \underbrace{\mathbb{E} \left[\frac{\eta}{2} \sum_{t=1}^T \sum_{i=1}^N p_{t,i} (\hat{\ell}_{t,i})^2 \right]}_C$$

Lower bound of A (using definition of loss estimates)

$$\mathbb{E} \left[\sum_{t=1}^T \sum_{i=1}^N p_{t,i} \hat{\ell}_{t,i} \right] \geq \mathbb{E} \left[\sum_{t=1}^T \sum_{i=1}^N p_{t,i} \ell_{t,i} \right] - \mathbb{E} \left[\gamma \sum_{t=1}^T \sum_{i=1}^N \frac{p_{t,i}}{\sigma_{t,i} + \gamma} \right]$$

Lower bound of B (optimistic loss estimates: $\mathbb{E}[\hat{\ell}] < \mathbb{E}[\ell]$)

$$-\mathbb{E} \left[\sum_{t=1}^T \hat{\ell}_{t,k} \right] \geq -\mathbb{E} \left[\sum_{t=1}^T \ell_{t,k} \right]$$

Upper bound of C (using definition of loss estimates)

$$\mathbb{E} \left[\frac{\eta}{2} \sum_{t=1}^T \sum_{i=1}^N p_{t,i} (\hat{\ell}_{t,i})^2 \right] \leq \mathbb{E} \left[\frac{\eta}{2} \sum_{t=1}^T \sum_{i=1}^N \frac{p_{t,i}}{\sigma_{t,i} + \gamma} \right]$$

Regret bound of Exp3-IX

$$R_T \leq \frac{\log N}{\eta} + \left(\frac{\eta}{2} + \gamma\right) \sum_{t=1}^T \mathbb{E} \left[\sum_{i=1}^N \frac{p_{t,i}}{o_{t,i} + \gamma} \right]$$

$$R_T \approx \mathcal{O} \left(\sqrt{\log N \sum_{t=1}^T \mathbb{E} \left[\sum_{i=1}^N \frac{p_{t,i}}{o_{t,i} + \gamma} \right]} \right)$$

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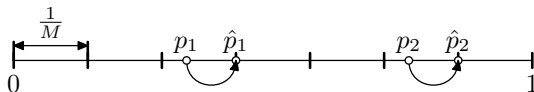
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Graph lemma

- ▶ Graph G with $V(G) = \{1, \dots, N\}$
- ▶ d_i^- – in-degree of vertex i
- ▶ α – independence set of G
- ▶ Turán's Theorem + induction

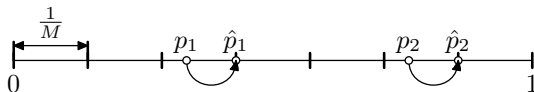
$$\sum_{i=1}^N \frac{1}{1 + d_i^-} \leq 2\alpha \log \left(1 + \frac{N}{\alpha} \right)$$

Discretization



$$\sum_{i=1}^N \frac{p_{t,i}}{o_{t,i} + \gamma} = \sum_{i=1}^N \frac{p_{t,i}}{p_{t,i} + \sum_{j \in N_i^-} p_{t,j} + \gamma} \leq \sum_{i=1}^N \frac{\hat{p}_{t,i}}{\hat{p}_{t,i} + \sum_{j \in N_i^-} \hat{p}_{t,j}} + 2$$

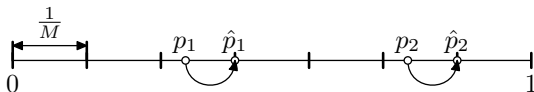
Discretization



$$\sum_{i=1}^N \frac{p_{t,i}}{o_{t,i} + \gamma} = \sum_{i=1}^N \frac{p_{t,i}}{p_{t,i} + \sum_{j \in N_i^-} p_{t,j} + \gamma} \leq \sum_{i=1}^N \frac{\hat{p}_{t,i}}{\hat{p}_{t,i} + \sum_{j \in N_i^-} \hat{p}_{t,j}} + 2$$

Note: we set $M = \lceil N^2/\gamma \rceil$

Discretization



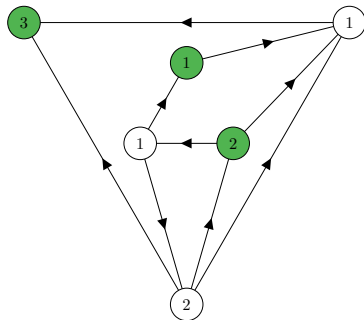
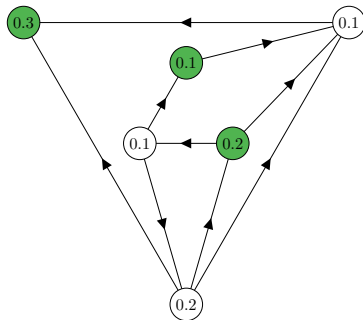
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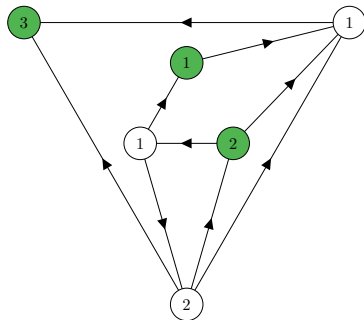
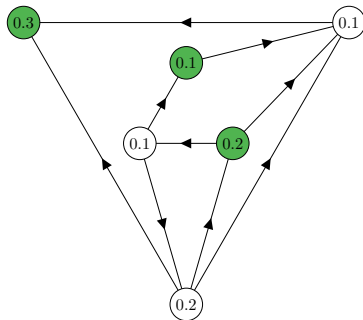
$$\sum_{i=1}^N \frac{\hat{p}_{t,i}}{\hat{p}_{t,i} + \sum_{j \in N_i^-} \hat{p}_{t,j}}$$

Example: let $M = 10$



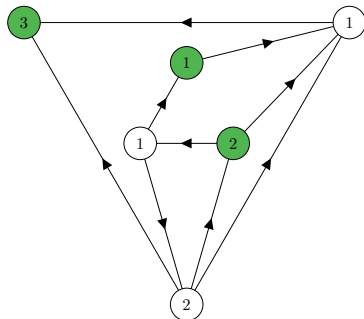
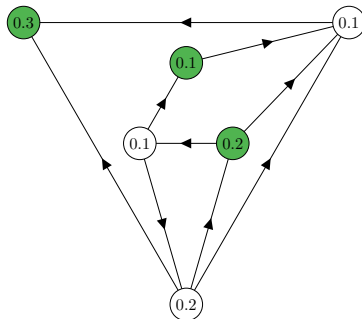
$$\sum_{i=1}^N \frac{M \hat{p}_{t,i}}{M \hat{p}_{t,i} + \sum_{j \in N_i^-} M \hat{p}_{t,j}}$$

Example: let $M = 10$



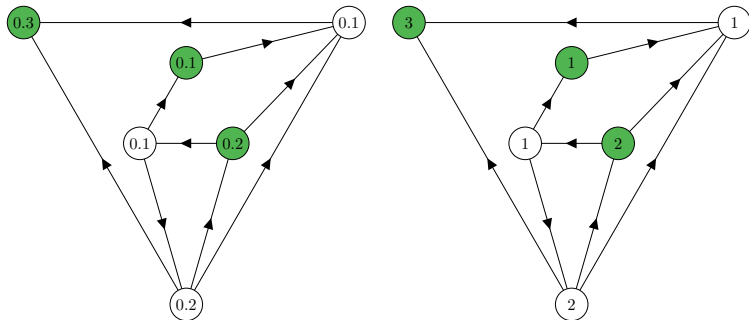
$$\sum_{i=1}^N \frac{M \hat{p}_{t,i}}{M \hat{p}_{t,i} + \sum_{j \in N_i^-} M \hat{p}_{t,j}} = \sum_{i=1}^N \sum_{k \in C_i} \frac{1}{1 + d_k^-}$$

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$$\sum_{i=1}^N \frac{M \hat{p}_{t,i}}{M \hat{p}_{t,i} + \sum_{j \in N_i^-} M \hat{p}_{t,j}} = \sum_{i=1}^N \sum_{k \in C_i} \frac{1}{1 + d_k^-} \leq 2\alpha \log \left(1 + \frac{M + N}{\alpha} \right)$$

Example: let $M = 10$



Exp3-IX regret bound

$$R_T \leq \frac{\log N}{\eta} + \left(\frac{\eta}{2} + \gamma\right) \sum_{t=1}^T \mathbb{E} \left[2\alpha_t \log \left(1 + \frac{\lceil N^2/\gamma \rceil + N}{\alpha_t} \right) + 2 \right]$$

$$R_T = \tilde{O} \left(\sqrt{\bar{\alpha} T \log(N)} \right)$$

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Next step

Exp3-IX regret bound

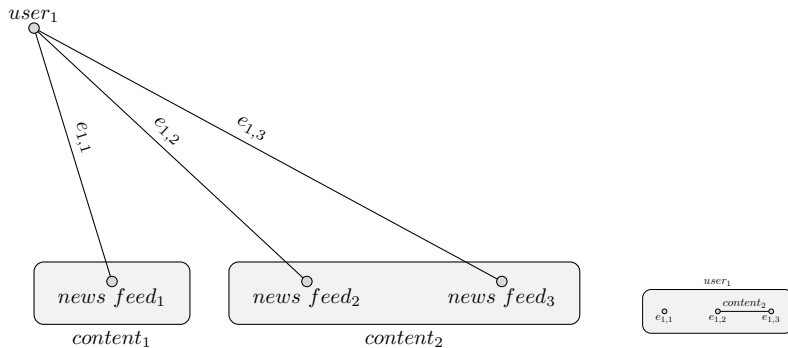
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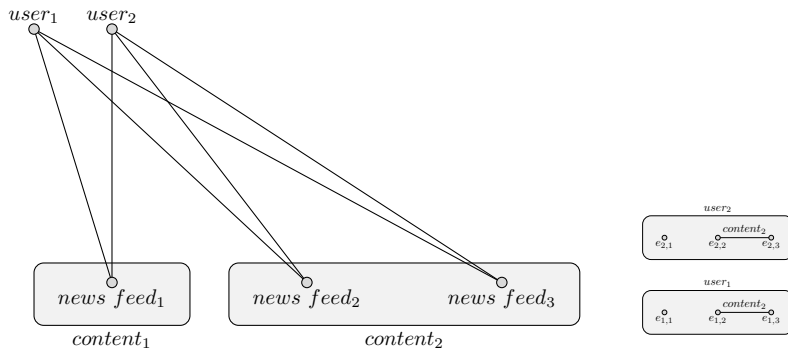
Next step

Generalization of the setting to **combinatorial actions**

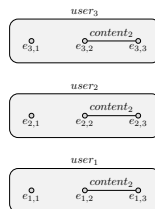
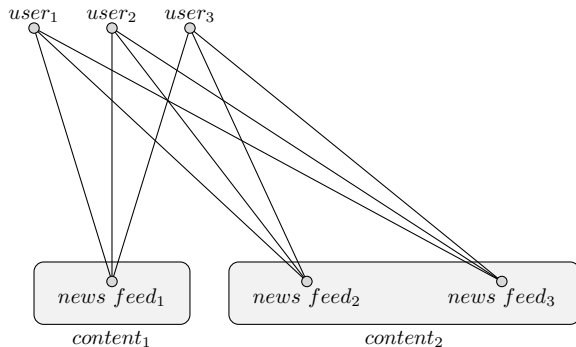
Example



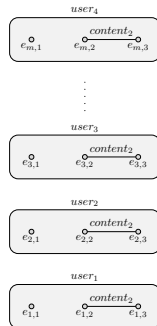
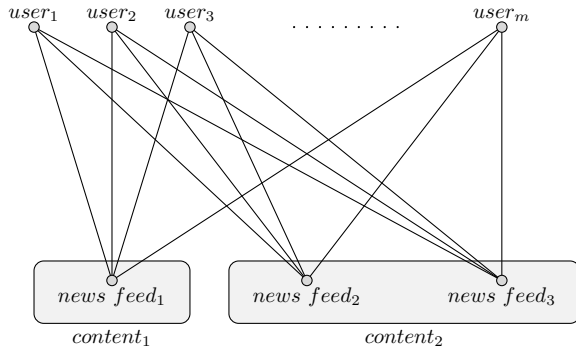
Example



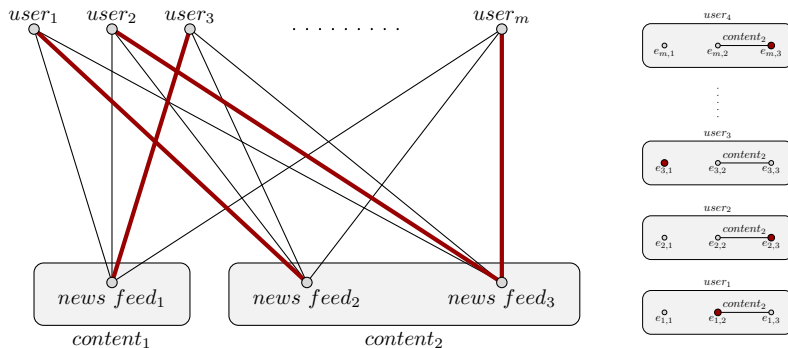
Example



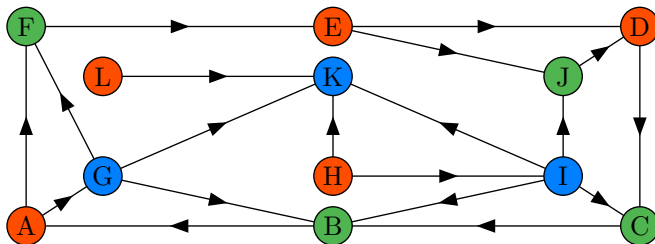
Example



Example



- ▶ Play m out of N nodes (combinatorial structure)
- ▶ Obtain losses of all played nodes
- ▶ Observe losses of all neighbors of played nodes



- ▶ Play action $\mathbf{V}_t \in S \subset \{0, 1\}^N$, $\|\mathbf{v}\|_1 \leq m$ for all $\mathbf{v} \in S$
- ▶ Obtain losses $\mathbf{V}_t^\top \ell_t$
- ▶ Observe additional losses according to the graph

FPL-IX algorithm

- ▶ Draw perturbation $Z_{t,i} \sim \text{Exp}(1)$ for all $i \in [N]$
- ▶ Play “the best” action $\mathbf{V}_t$ according to total loss estimate $\hat{\mathbf{L}}_{t-1}$ and perturbation $\mathbf{Z}_t$

$$\mathbf{V}_t = \arg \min_{\mathbf{v} \in \mathcal{S}} \mathbf{v}^\top \left(\eta_t \hat{\mathbf{L}}_{t-1} - \mathbf{Z}_t \right)$$

- ▶ Compute loss estimates

$$\hat{\ell}_{t,i} = \ell_{t,i} K_{t,i} \mathbb{1}\{\ell_{t,i} \text{ is observed}\}$$

- ▶ $K_{t,i}$: geometric random variable with

$$\mathbb{E}[K_{t,i}] = \frac{1}{o_{t,i} + (1 - o_{t,i})\gamma}$$

FPL-IX - regret bound

$$R_T = \tilde{O} \left(m^{3/2} \sqrt{\sum_{t=1}^T \alpha_t} \right) = \tilde{O} \left(m^{3/2} \sqrt{\bar{\alpha} T} \right)$$

Side Observation Summary

- ▶ **Implicit eXploration** idea
- ▶ **New algorithm for simple actions - Exp3-IX**
 - ▶ Using implicit exploration idea
 - ▶ Same regret bound as previous algorithm
 - ▶ No need to know graph before an action is played
 - ▶ Computationally efficient
- ▶ **New combinatorial setting with side observations**
- ▶ **Algorithm for combinatorial setting - FPL-IX**
- ▶ **Future directions**
 - ▶ No need to know graph after an action is played
 - ▶ Stochastic side observations
 - ▶ Random graph models
 - ▶ Exploiting the communities



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Sylvester's determinant theorem:

$$|\mathbf{A} + \mathbf{x}\mathbf{x}^\top| = |\mathbf{A}| |\mathbf{I} + \mathbf{A}^{-1}\mathbf{x}\mathbf{x}^\top| = |\mathbf{A}| (1 + \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x})$$

Goal:

- ▶ Upperbound determinant $|\mathbf{A} + \mathbf{x}\mathbf{x}^\top|$ for $\|\mathbf{x}\|_2 \leq 1$
- ▶ Upperbound $\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x}$

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$$\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} = \mathbf{x}^\top \mathbf{Q} \mathbf{\Lambda}^{-1} \mathbf{Q}^\top \mathbf{x} = \mathbf{y}^\top \mathbf{\Lambda}^{-1} \mathbf{y} = \sum_{i=1}^N \lambda_i y_i^2$$

Sylvester's determinant theorem:

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- ▶ $\|\mathbf{y}\|_2 \leq 1$.
- ▶ $\mathbf{y}$ is a canonical vector.
- ▶ $\mathbf{x} = \mathbf{Q}\mathbf{y}$ is an eigenvector of $\mathbf{A}$.

Corollary:

Determinant $|\mathbf{V}_T|$ of $\mathbf{V}_T = \mathbf{\Lambda} + \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^\top$ is maximized when all $\mathbf{x}_t$ are aligned with axes.

$$\begin{aligned}
 |\mathbf{V}_T| &\leq \max_{\sum t_i = T} \prod (\lambda_i + t_i) \\
 \ln \frac{|\mathbf{V}_T|}{|\mathbf{\Lambda}|} &\leq \max_{\sum t_i = T} \sum \ln \left(1 + \frac{t_i}{\lambda_i} \right) \\
 \ln \frac{|\mathbf{V}_T|}{|\mathbf{\Lambda}|} &\leq \sum_{i=1}^d \ln \left(1 + \frac{T}{\lambda} \right) + \sum_{i=d+1}^N \ln \left(1 + \frac{t_i}{\lambda_{d+1}} \right) \\
 &\leq d \ln \left(1 + \frac{T}{\lambda} \right) + \frac{T}{\lambda_{d+1}} \\
 &\leq 2d \ln \left(1 + \frac{T}{\lambda} \right)
 \end{aligned}$$

$$\mathbf{f}^\top \mathcal{L} \mathbf{f} = \frac{1}{2} \sum_{i,j \leq N} w_{i,j} (f_i - f_j)^2 = S_G(\mathbf{f})$$

Proof:

$$\begin{aligned} \mathbf{f}^\top \mathcal{L} \mathbf{f} &= \mathbf{f}^\top \mathcal{D} \mathbf{f} - \mathbf{f}^\top \mathcal{W} \mathbf{f} = \sum_{i=1}^N d_i f_i^2 - \sum_{i,j \leq N} w_{i,j} f_i f_j \\ &= \frac{1}{2} \left(\sum_{i=1}^N d_i f_i^2 - 2 \sum_{i,j \leq N} w_{i,j} f_i f_j + \sum_{j=1}^N d_j f_j^2 \right) = \frac{1}{2} \sum_{i,j \leq N} w_{i,j} (f_i - f_j)^2 \end{aligned}$$

SpectralUCB analysis sketch

- Derivation of the confidence ellipsoid for $\hat{\alpha}$ with probability $1 - \delta$.
 - Using analysis of OFUL (Abbasi-Yadkori et al., 2011)

$$|\mathbf{x}^\top(\hat{\alpha} - \alpha^*)| \leq \|\mathbf{x}\|_{\mathbf{V}_t^{-1}} \left(R \sqrt{2 \ln \left(\frac{|\mathbf{V}_t|^{1/2}}{\delta |\mathbf{\Lambda}|^{1/2}} \right)} + C \right)$$

- Regret in one time step: $r_t = \mathbf{x}_*^\top \alpha^* - \mathbf{x}_{\pi(t)}^\top \alpha^* \leq 2c_t \|\mathbf{x}_{\pi(t)}\|_{\mathbf{V}_t^{-1}}$
- Cumulative regret:

$$R_T = \sum_{t=1}^T r_t \leq \sqrt{T \sum_{t=1}^T r_t^2} \leq 2(c_T + 1) \sqrt{2T \ln \frac{|\mathbf{V}_T|}{|\mathbf{\Lambda}|}}$$

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- Derivation of the confidence ellipsoid for $\hat{\alpha}$ with probability $1 - \delta$.
 - Using analysis of OFUL (Abbasi-Yadkori et al., 2011)

$$|\mathbf{x}^\top(\hat{\alpha} - \alpha^*)| \leq \|\mathbf{x}\|_{\mathbf{V}_t^{-1}} \left(R \sqrt{2 \ln \left(\frac{|\mathbf{V}_t|^{1/2}}{\delta |\mathbf{\Lambda}|^{1/2}} \right)} + C \right)$$

- Regret in one time step: $r_t = \mathbf{x}_*^\top \alpha^* - \mathbf{x}_{\pi(t)}^\top \alpha^* \leq 2c_t \|\mathbf{x}_{\pi(t)}\|_{\mathbf{V}_t^{-1}}$
- Cumulative regret:

$$R_T = \sum_{t=1}^T r_t \leq \sqrt{T \sum_{t=1}^T r_t^2} \leq 2(\underbrace{c_T}_{\text{red}} + 1) \sqrt{2T \ln \frac{|\mathbf{V}_T|}{|\mathbf{\Lambda}|}}$$

- Upperbound for $\ln(|\mathbf{V}_t|/|\mathbf{\Lambda}|)$

$$\ln \frac{|\mathbf{V}_t|}{|\mathbf{\Lambda}|} \leq \ln \frac{|\mathbf{V}_T|}{|\mathbf{\Lambda}|} \leq 2d \ln \left(\frac{\lambda + T}{\lambda} \right)$$

SpectralTS analysis sketch

Divide arms into two groups

- ▶ $\Delta_i = \mathbf{b}_*^\top \boldsymbol{\mu} - \mathbf{b}_i^\top \boldsymbol{\mu} \leq g \|\mathbf{b}_i\|_{\mathbf{B}_t^{-1}}$ arm i is **unsaturated**
- ▶ $\Delta_i = \mathbf{b}_*^\top \boldsymbol{\mu} - \mathbf{b}_i^\top \boldsymbol{\mu} > g \|\mathbf{b}_i\|_{\mathbf{B}_t^{-1}}$ arm i is **saturated**

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Saturated arm

- ▶ Small standard deviation $\rightarrow$ accurate regret estimate.
- ▶ **High regret** on playing the arm $\rightarrow$ **Low probability** of picking

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Unsaturated arm

- ▶ **Low regret** bounded by a factor of standard deviation
- ▶ **High probability** of picking

SpectralTS analysis sketch

- Confidence ellipsoid for estimate $\hat{\boldsymbol{\mu}}$ of $\boldsymbol{\mu}$ (with probability $1 - \delta/T^2$)
 - Using analysis of OFUL algorithm (Abbasi-Yadkori et al., 2011)

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- Our key result coming from spectral properties of $\mathbf{B}_t$.

$$\log \frac{|\mathbf{B}_t|}{|\boldsymbol{\Lambda}|} \leq 2d \log \left(1 + \frac{T}{\lambda} \right)$$

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- Concentration of sample $\tilde{\boldsymbol{\mu}}$ around mean $\hat{\boldsymbol{\mu}}$ (with probability $1 - 1/T^2$)
 - Using concentration inequality for Gaussian random variable.

$$|\mathbf{b}_i^\top \tilde{\boldsymbol{\mu}} - \mathbf{b}_i^\top \hat{\boldsymbol{\mu}}| \leq \left(R \sqrt{6d \log \left(\frac{\lambda + T}{\delta\lambda} \right)} + C \right) \|\mathbf{b}_i\|_{\mathbf{B}_t^{-1}} \sqrt{4 \log(TN)} = v \|\mathbf{b}_i\|_{\mathbf{B}_t^{-1}} \sqrt{4 \log(TN)}$$

SpectralTS analysis sketch

Define $\text{regret}'(t) = \text{regret}(t) \cdot \mathbb{1}\{|\mathbf{b}_i^\top \hat{\boldsymbol{\mu}}(t) - \mathbf{b}_i^\top \boldsymbol{\mu}| \leq \ell \|\mathbf{b}_i\|_{\mathbf{B}_t^{-1}}\}$

$$\text{regret}'(t) \leq \frac{11g}{p} \|\mathbf{b}_{a(t)}\|_{\mathbf{B}_t^{-1}} + \frac{1}{T^2}$$

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Super-martingale (i.e. $\mathbb{E}[Y_t - Y_{t-1} | \mathcal{F}_{t-1}] \leq 0$)

$$X_t = \text{regret}'(t) - \frac{11g}{p} \|\mathbf{b}_{a(t)}\|_{\mathbf{B}_t^{-1}} - \frac{1}{T^2}$$

$$Y_t = \sum_{w=1}^t X_w.$$

$(Y_t; t = 0, \dots, T)$ is a **super-martingale** process w.r.t. history $\mathcal{F}_t$.

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Backup: SpectralEliminator pseudocode

Input:

N : the number of nodes, T : the number of pulls

$\{\Lambda_{\mathcal{L}}, \mathbf{Q}\}$ spectral basis of $\mathcal{L}$

λ : regularization parameter

$\beta, \{t_j\}_j^J$ parameters of the elimination and phases

$A_1 \leftarrow \{\mathbf{x}_1, \dots, \mathbf{x}_K\}$.

for $j = 1$ **to** J **do**

$\mathbf{V}_{t_j} \leftarrow \gamma \Lambda_{\mathcal{L}} + \lambda \mathbf{I}$

for $t = t_j$ **to** $\min(t_{j+1} - 1, T)$ **do**

Play $\mathbf{x}_t \in A_j$ with the largest width to observe r_t :

$\mathbf{x}_t \leftarrow \arg \max_{\mathbf{x} \in A_j} \|\mathbf{x}\|_{\mathbf{V}_t^{-1}}$

$\mathbf{V}_{t+1} \leftarrow \mathbf{V}_t + \mathbf{x}_t \mathbf{x}_t^\top$

end for

Eliminate the arms that are not promising:

$\hat{\alpha}_t \leftarrow \mathbf{V}_t^{-1} [\mathbf{x}_{t_j}, \dots, \mathbf{x}_t] [r_{t_j}, \dots, r_t]^\top$

$A_{j+1} \leftarrow \left\{ \mathbf{x} \in A_j, \langle \hat{\alpha}_t, \mathbf{x} \rangle + \|\mathbf{x}\|_{\mathbf{V}_t^{-1}} \beta \geq \max_{\mathbf{x} \in A_j} \left[\langle \hat{\alpha}_t, \mathbf{x} \rangle - \|\mathbf{x}\|_{\mathbf{V}_t^{-1}} \beta \right] \right\}$

end for

Backup: SpectralEliminator analysis

SpectralEliminator

- ▶ Divide time into sets ($t_1 = 1 \leq t_2 \leq \dots$) to introduce independence for Azuma-Hoeffding inequality and observe

$$R_T \leq \sum_{j=0}^J (t_{j+1} - t_j) [\langle \mathbf{x}^* - \mathbf{x}_t, \hat{\alpha}_j \rangle + (\|\mathbf{x}^*\|_{\mathbf{V}_j^{-1}} + \|\mathbf{x}_t\|_{\mathbf{V}_j^{-1}}) \beta]$$
- ▶ Bound $\langle \mathbf{x}^* - \mathbf{x}_t, \hat{\alpha}_j \rangle$ for each phase
- ▶ No bad arms: $\langle \mathbf{x}^* - \mathbf{x}_t, \hat{\alpha}_j \rangle \leq (\|\mathbf{x}^*\|_{\mathbf{V}_j^{-1}} + \|\mathbf{x}_t\|_{\mathbf{V}_j^{-1}}) \beta$
- ▶ By algorithm: $\|\mathbf{x}\|_{\mathbf{V}_j^{-1}}^2 \leq \frac{1}{t_j - t_{j-1}} \sum_{s=t_{j-1}+1}^{t_j} \|\mathbf{x}_s\|_{\mathbf{V}_{s-1}^{-1}}^2$
- ▶ $\sum_{s=t_{j-1}+1}^{t_j} \min \left(1, \|\mathbf{x}_s\|_{\mathbf{V}_{s-1}^{-1}}^2 \right) \leq \log \frac{|\mathbf{V}_j|}{|\Lambda|}$